COMPUTER SCIENCE & IT

DIGITAL LOGIC

Lecture No. 03

Combinational Circuit

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F.A.

Serial adder & Parallel adder

H.S. & F.S.

- To add two n-bit numbers $\rightarrow$ How many HA's and OR gates required ?

$1\ H.A. + (n-1)\ F.A.$

$1\ H.A. + 2(n-1)\ HA + (n-1)\ OR$

$(2n-1)\ H.A. + (n-1)\ OR$

[Question 4]

To add two 4-bit numbers, minimum number of H.A.s require is m and minimum number of OR gates require is n then value of $(m + n)$ is __10__.

$1HA + 3F.A$

$= 1HA + 6HA + 3OR$

$= 7HA + 3OR$

$m = 7HA$

$n = 3OR$

$m + n = 10$

$$\begin{array}{rccc} & 7 & 8 & 6 \\ & 4 & 5 & 8 \\ - & 0 & 1 & \\ \hline 0 & 3 & 2 & 8 \\ \hline \end{array}$$

$$\begin{array}{lcccc} & 1 & 0 & 1 & 0 \\ \text{Result } +ve & 0 & 1 & 1 & 1 \\ - & 1 & 1 & 1 & \\ \hline 0 & 0 & 0 & 1 & 1 \\ \hline \end{array}$$

$$\begin{array}{lccccc} & & 0 & 1 & 1 & 1 \\ \text{Result } -ve & & 1 & 0 & 1 & 0 \\ & & 0 & 0 & 0 & \\ \hline \text{Result} & 1 & 1 & 1 & 0 & 1 \\ \hline \end{array}$$

2's

$-(00011)$

$$
\begin{array}{ccccc}
 & 1 & 0 & 1 & 0 \\
 & 0 & 0 & 0 & 1 \\
- & 0 & 0 & 1 & \\
\hline
0 & 1 & 0 & 0 & 1 \\
\hline
\end{array}
$$

$+9$

$$
\begin{array}{ccccc}
 & 0 & 0 & 0 & 1 \\
 & 1 & 0 & 1 & 0 \\
 & 1 & 1 & 0 & \\
\hline
1 & 0 & 1 & 1 & 1 \\
\hline
\end{array}
\longrightarrow \text{Result}
$$

Result -ve

$-(2^{s}\text{ compliment } 10111)$

$-(\ 01001)$

$-(9)$

$$
\begin{array}{lccccc}
A = & & a_3 & a_2 & a_1 & a_0 \\
B = & & b_3 & b_2 & b_1 & b_0 \\
 & - & B_3 & B_2 & B_1 & \\
\hline
 & B_4 & D_3 & D_2 & D_1 & D_0 \\
\hline
\end{array}
$$

Half Subtractor

- What is Half subtractor and what are input lines and what are output lines ?

- I/P lines ⟶ 2-i/P lines ⟶ x, y ⟶ corresponding bits of the two no.s

  $x \to a_0$, $y \to b_0$; $x \to a_1$, $y \to b_1$; $x \to a_2$, $y \to b_2$ etc.

- 2-O/P lines ⟶ D ⟶ Difference O/P ⟶ after considering borrow, the result of subtraction is difference O/P.

  B – Borrow O/P ⟶ Borrow O/P will be '1' if borrow is needed for $x-y$ operation otherwise '0'.

$x - y$

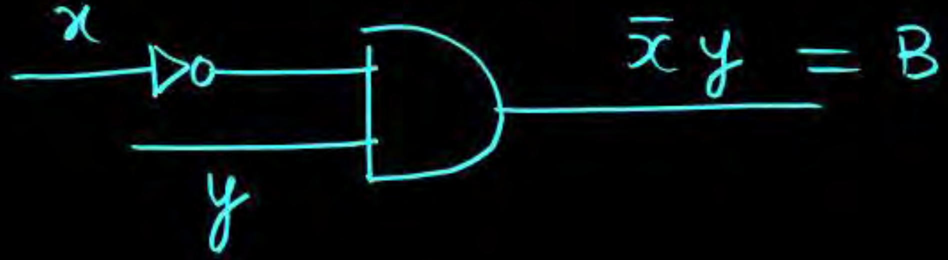
	Input lines		Output lines	
	x	y	D-o/p	B-o/p
0	0	0	0	0
1	0	1	1	1
2	1	0	1	0
3	1	1	0	0

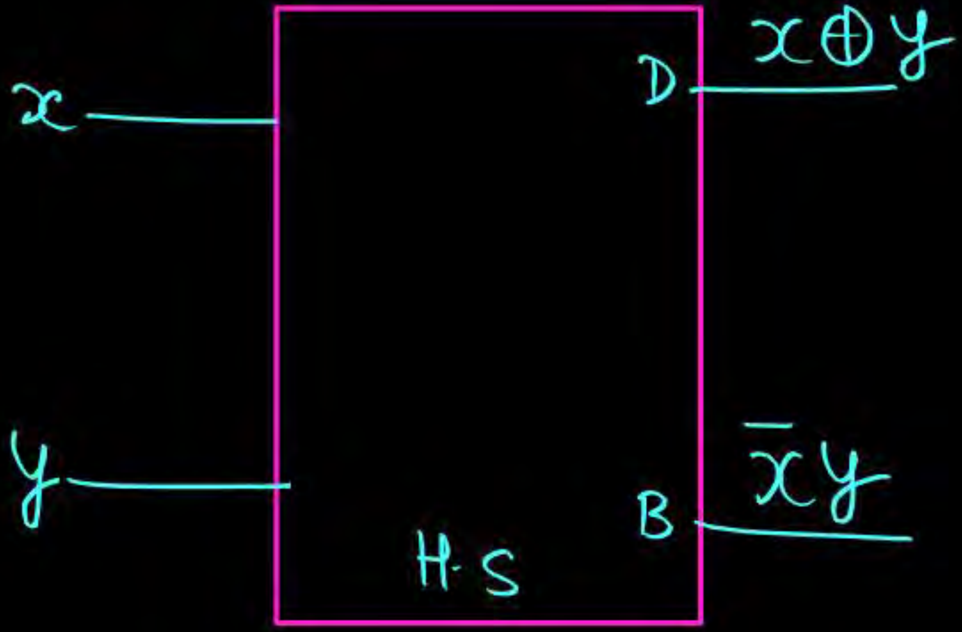
$D(x,y) = \sum(1,2) = \bar{x}y + x\bar{y} = x \oplus y$

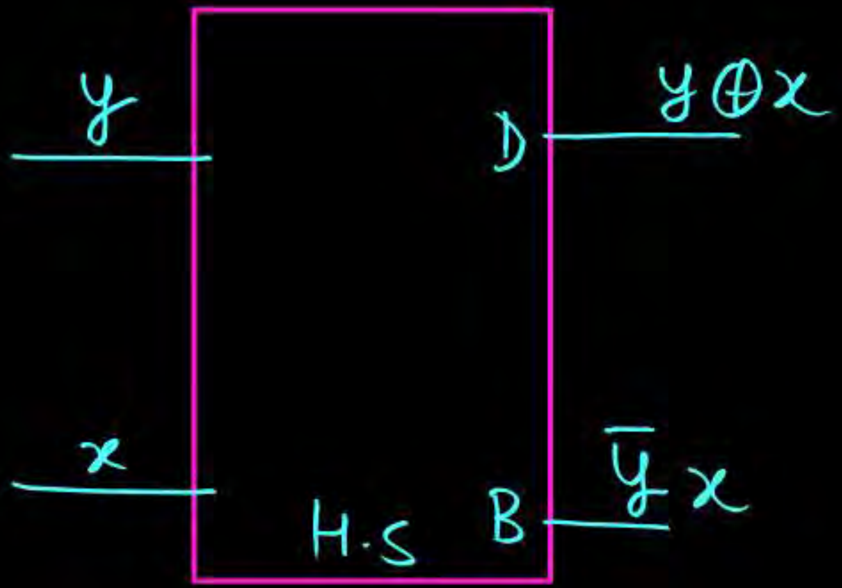
→ Non self dual

$B(x,y) = \sum(1) = \bar{x} \cdot y$

→ Non-self dual

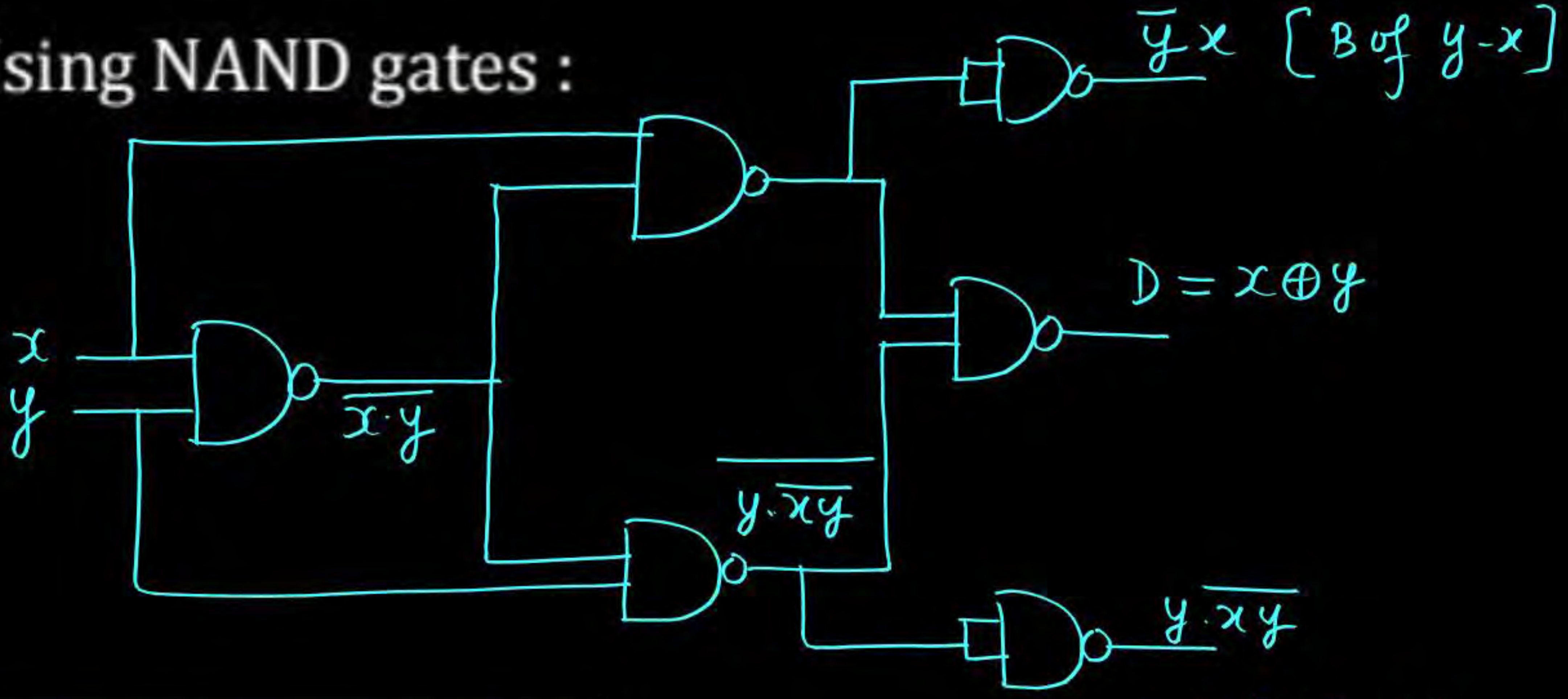






- Using NAND gates :

$x - y$



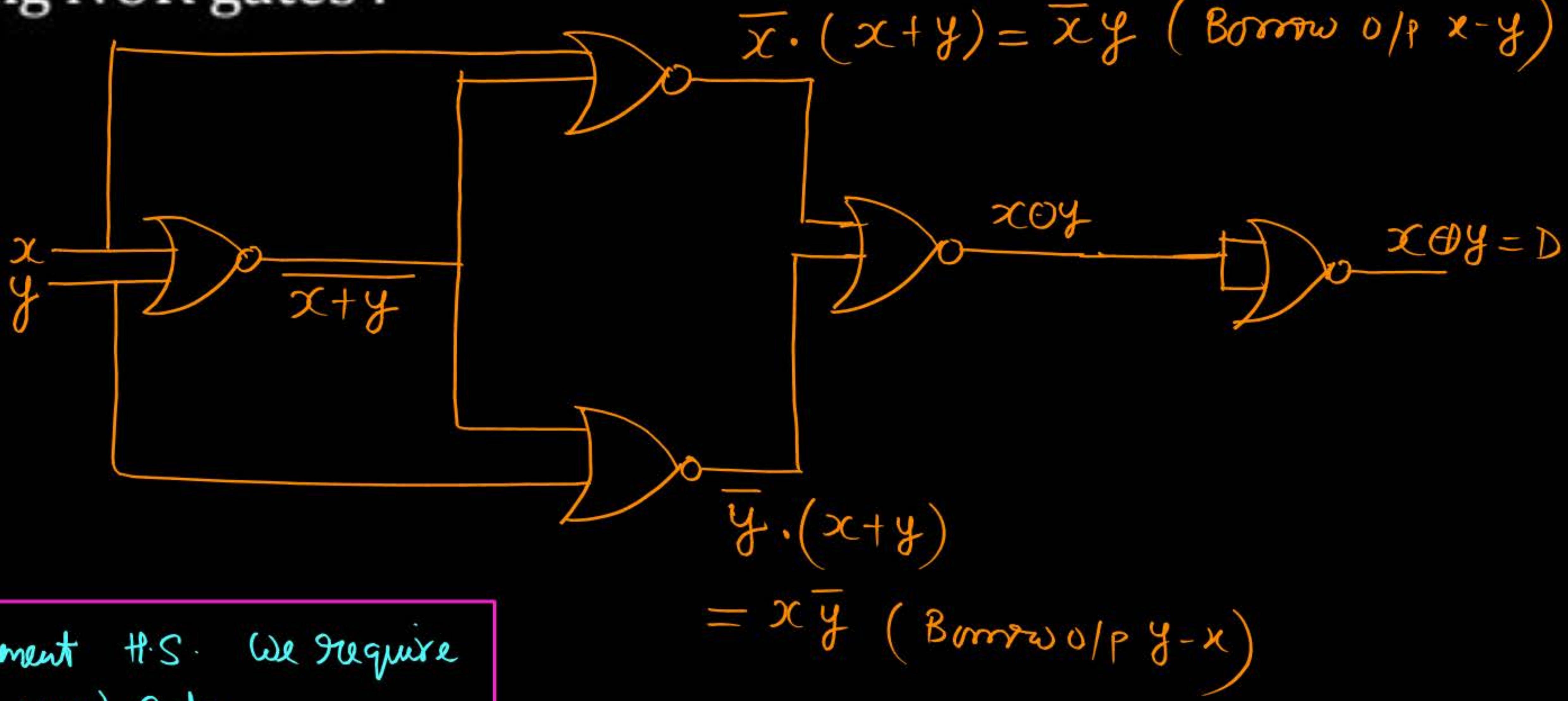
$= y(\bar{x} + \bar{y})$

$= \bar{x}y$

B of $(x-y)$

- To implement H.S. we require 5 [2-i/p] NAND gates.

- Using NOR gates :



- To implement H.S. we require 5 (2-i/p NOR) gates.

Full Subtractor

- What is full subtractor and what are input lines and what are output lines ?

→ I/P lines → 3-I/P lines → x, y → corresponding bits of the two numbers.
z → borrow O/P from previous subtraction.

x, y → a_1, b_1 then $z \rightarrow B_1$ (borrow O/P from a_0, b_0 position)
→ a_2, b_2 then $z \rightarrow B_2$ (borrow O/P from a_1, b_1 position)

→ O/P lines → Difference O/P D → after considering borrow, the result of subtraction is difference O/P D.

Borrow O/P → will be '1' if borrow is needed for $x-y-z$ otherwise '0'.

$x-y-z$

	Input lines			Output lines	
	x	y	z	D-o/p	B-o/p
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

$$D(x,y,z) = \sum(1,2,4,7) = \pi(0,3,5,6)$$

$$= (x \oplus y \oplus z)$$

Self dual

$$B(x,y,z) = \sum(1,2,3,7) = \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz$$

$$= \bar{x}y + yz + \bar{x}z$$

Self dual

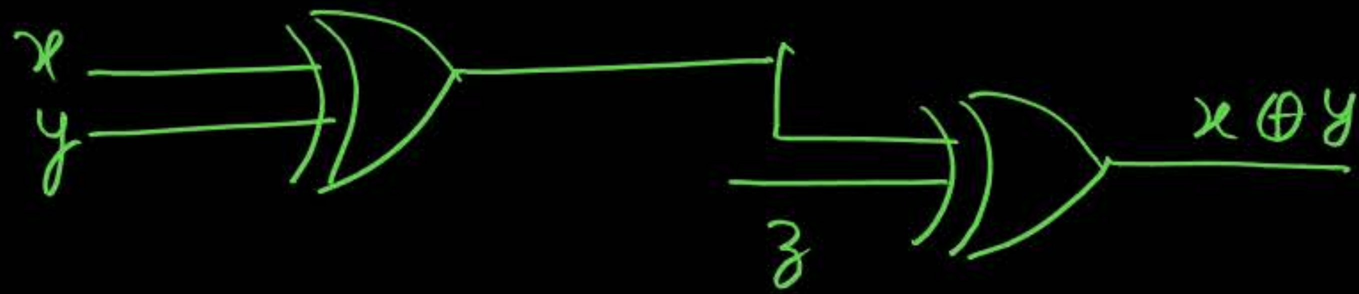
$$= \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz$$

$$= \bar{x}y + z[\bar{x}\bar{y} + xy] = \bar{x}y + z[\bar{x} \oplus y]$$

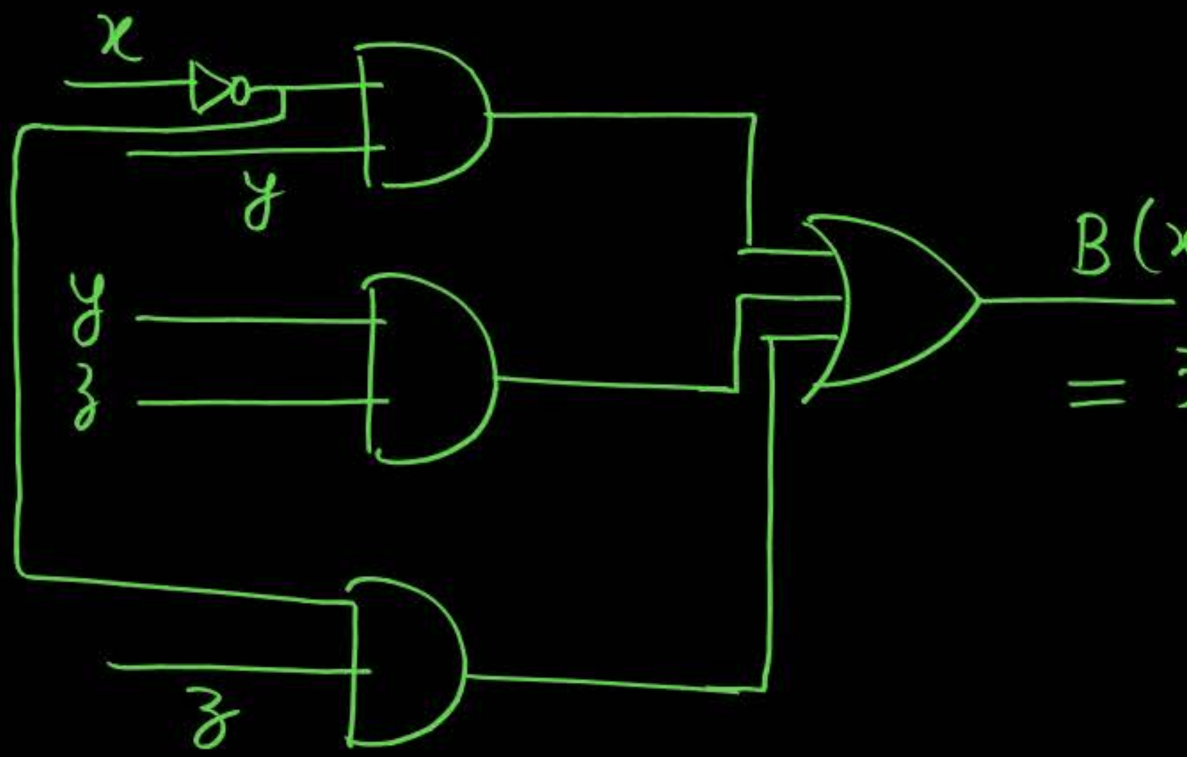
$$y-x-z \longrightarrow (x\bar{y} + \bar{y}z + xz) = B$$

$$x+y \longrightarrow xy \qquad x+y+z \longrightarrow xy+yz+zx, \quad xy + z[x \oplus y]$$

$$x-y \longrightarrow \bar{x}y \qquad x-y-z \longrightarrow \bar{x}y+yz+z\bar{x}, \quad \bar{x}y + z[\bar{x} \oplus y]$$



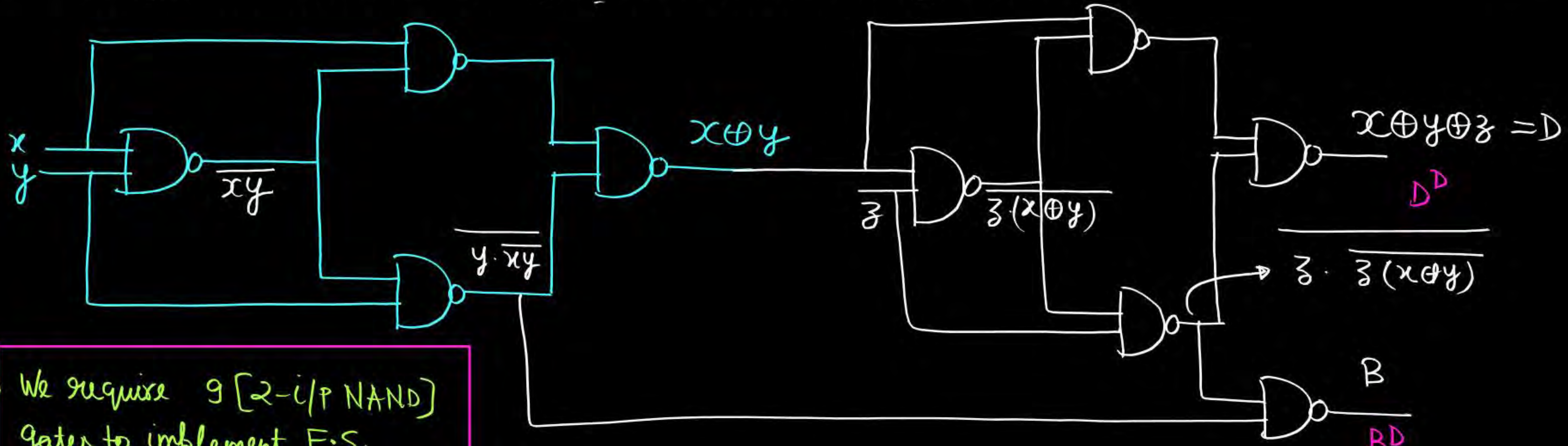
$x \oplus y \oplus z = D(x, y, z)$



$B(x, y, z)$

$= \bar{x}y + yz + \bar{x}z$

- Implementation using NAND and NOR gates :



- We require 9 [2-i/p NAND] gates to implement F.S.

NOR → H.W.
(9)

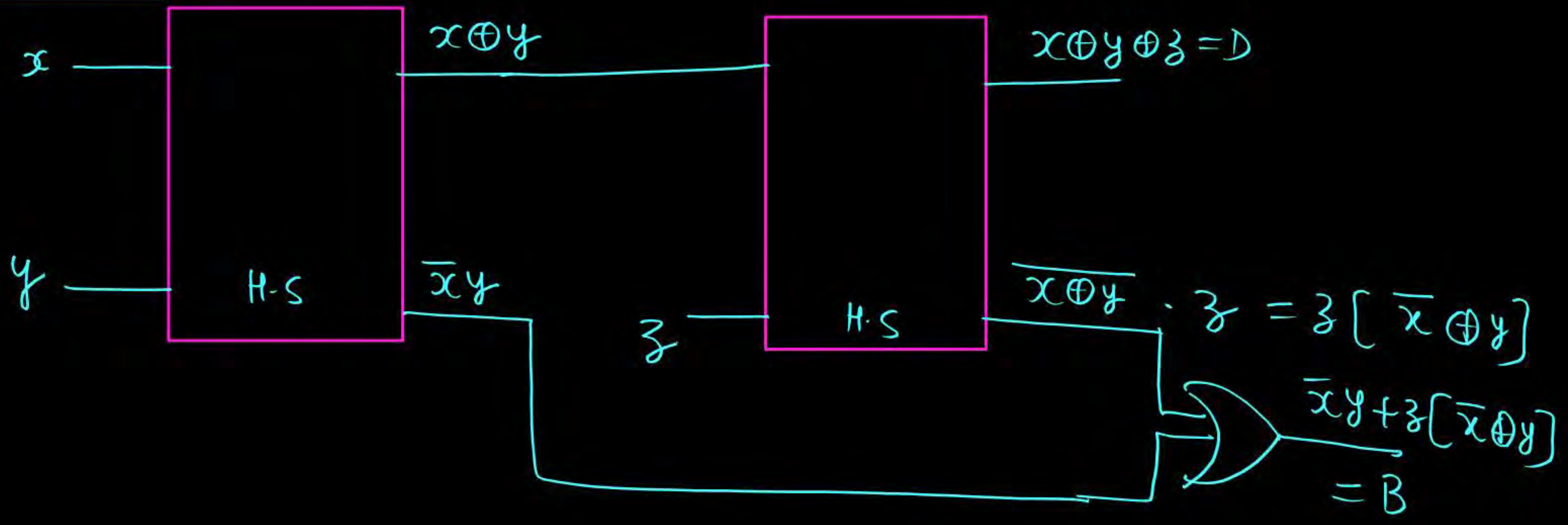
$$= y\cdot\overline{xy} + z\cdot\overline{z\cdot(x\oplus y)}$$
$$= \bar{x}y + z\cdot[\bar{z} + x\odot y]$$
$$= \bar{x}y + z[\bar{x}\oplus y]$$

- Implementation using H.S and OR gate :

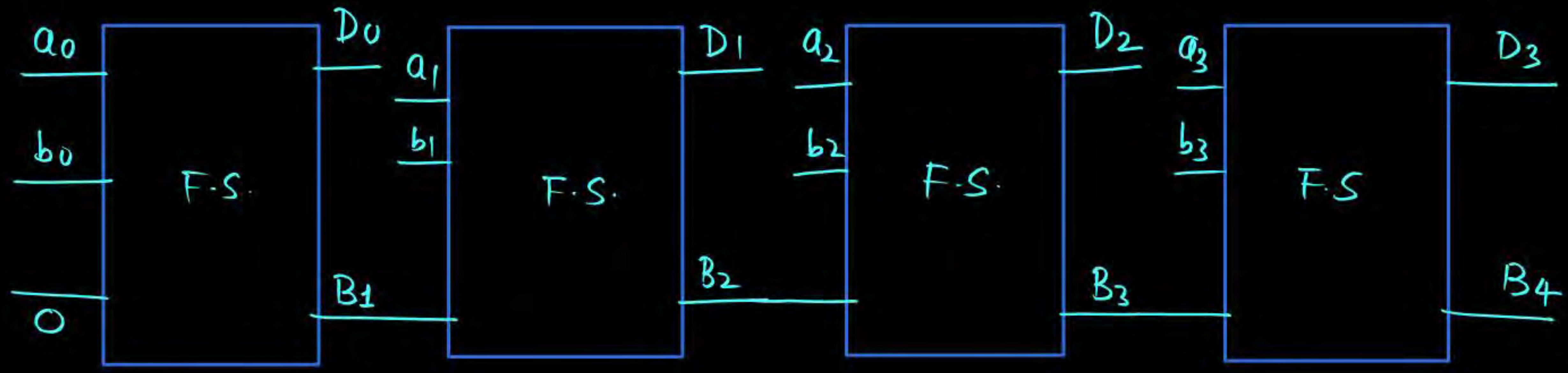
$(x-y-z) \longrightarrow D = x \oplus y \oplus z$

$B = \bar{x}y + yz + \bar{x}z = \bar{x}y + z[\overline{x \oplus y}]$

$\boxed{1 FS \rightarrow 2 H.S + 1 OR}$



Parallel Subtractor



$$\text{Result} = (B_4 D_3 D_2 D_1 D_0)$$

- To subtract two n-bit numbers,, number of H.S. and number of OR gate required is :

$$\boxed{\begin{array}{l} 1\,H.S. + (n-1)\,F.S. \\ 1\,H.S + 2(n-1)\,H.S + (n-1)\,OR \\ (2n-1)\,H.S + (n-1)\,OR. \end{array}}$$

[Question]

Two subtract two 4-bit numbers, minimum number of H.S. required is m and minimum number of OR gates require is n then value of $(m + n)$ is ___10___.

1 H.S + 3 F.S.

1 H.S + 6 H.S + 3 OR

= 7 H.S + 3 OR

$m = 7$

$n = 3$

$m + n = 10.$

$x+y+z \longrightarrow \text{Carry} = xy+yz+zx$

$= xy + (x\oplus y)z$

$y+z+x \longrightarrow \text{Carry} = yz + (y\oplus z)x$

$z+x+y \longrightarrow \text{Carry} = zx + (z\oplus x)\cdot y$

$\longrightarrow x-y-z \longrightarrow B = \bar{x}y + yz + \bar{x}z$

$= \bar{x}y + (\bar{x}\oplus y)z$

$= yz + (y\oplus z)\bar{x}$

$= \bar{x}z + (z\oplus \bar{x})\cdot y$

$\longrightarrow y-x-z = x\bar{y} + \bar{y}z + xz$

$= x\bar{y} + (x\oplus \bar{y})z$

$= \bar{y}z + (\bar{y}\oplus z)x$

$= xz + (z\oplus x)\bar{y}$

$\longrightarrow z-x-y \longrightarrow xy + y\bar{z} + \bar{z}x$ $\quad \bar{z}x + (\bar{z}\oplus x)y$

$\longrightarrow xy + (x\oplus y)\bar{z}$

$\longrightarrow y\bar{z} + (y\oplus \bar{z})x$

2 Minute Summary

$\boxed{\text{H.S. \& F.S.}}$

Thank you

GW

Soldiers !